

```
In [1]: import pandas as pd
```

```
In [2]: # Create sample data
data = {
    'ID': [101, 102, 103, 104, 105, 105, 106, 107],
    'Name': ['John', 'Alice', 'Bob', 'John', None, 'Bob', 'Eve', 'ALICE'],
    'Age': [25, 30, None, 22, 28, 28, None, 30],
    'City': ['New York', 'Los Angeles', 'New York', None, 'Chicago', 'Chicago', 'Miami', 'los angeles']
}
```

```
In [3]: # Create DataFrame
df = pd.DataFrame(data)
print("Original Dataset:\n", df)
```

Original Dataset:

	ID	Name	Age	City
0	101	John	25.0	New York
1	102	Alice	30.0	Los Angeles
2	103	Bob	NaN	New York
3	104	John	22.0	None
4	105	None	28.0	Chicago
5	105	Bob	28.0	Chicago
6	106	Eve	NaN	Miami
7	107	ALICE	30.0	los angeles

```
In [4]: print("\nChecking for Missing Values:")
print(df.isnull().sum())
```

Checking for Missing Values:

```
ID      0
Name     1
Age      2
City     1
dtype: int64
```

```
In [5]: df_dropped = df.dropna()
print("\nAfter Dropping Missing Values:\n", df_dropped)
```

After Dropping Missing Values:

	ID	Name	Age	City
0	101	John	25.0	New York
1	102	Alice	30.0	Los Angeles
5	105	Bob	28.0	Chicago
7	107	ALICE	30.0	los angeles

```
In [6]: # Fill missing names with 'Unknown' and ages with mean age
df['Name'].fillna('Unknown', inplace=True)
df['Age'].fillna(df['Age'].mean(), inplace=True)
df['City'].fillna('Unknown', inplace=True)

print("\nAfter Filling Missing Values:\n", df)
```

After Filling Missing Values:

	ID	Name	Age	City
0	101	John	25.000000	New York
1	102	Alice	30.000000	Los Angeles
2	103	Bob	27.166667	New York
3	104	John	22.000000	Unknown
4	105	Unknown	28.000000	Chicago
5	105	Bob	28.000000	Chicago
6	106	Eve	27.166667	Miami
7	107	ALICE	30.000000	los angeles

```
In [7]: print("\nDuplicate Rows:")
print(df[df.duplicated()])
```

Duplicate Rows:
Empty DataFrame
Columns: [ID, Name, Age, City]
Index: []

```
In [8]: df = df.drop_duplicates()
print("\nAfter Removing Duplicates:\n", df)
```

After Removing Duplicates:

	ID	Name	Age	City
0	101	John	25.000000	New York
1	102	Alice	30.000000	Los Angeles
2	103	Bob	27.166667	New York
3	104	John	22.000000	Unknown
4	105	Unknown	28.000000	Chicago
5	105	Bob	28.000000	Chicago
6	106	Eve	27.166667	Miami
7	107	ALICE	30.000000	los angeles

```
In [9]: # Convert all city names and names to title case
df['Name'] = df['Name'].str.title()
df['City'] = df['City'].str.title()

print("\nAfter Correcting Text Case:\n", df)
```

After Correcting Text Case:

	ID	Name	Age	City
0	101	John	25.000000	New York
1	102	Alice	30.000000	Los Angeles
2	103	Bob	27.166667	New York
3	104	John	22.000000	Unknown
4	105	Unknown	28.000000	Chicago
5	105	Bob	28.000000	Chicago
6	106	Eve	27.166667	Miami
7	107	Alice	30.000000	Los Angeles

```
In [10]: print("\nUnique Names:", df['Name'].unique())
print("\nUnique Cities:", df['City'].unique())
```

```
Unique Names: ['John' 'Alice' 'Bob' 'Unknown' 'Eve']
Unique Cities: ['New York' 'Los Angeles' 'Unknown' 'Chicago' 'Miami']
```

```
In [11]: print("\n✅ Final Cleaned Dataset:\n", df)
```



Final Cleaned Dataset:

	ID	Name	Age	City
0	101	John	25.000000	New York
1	102	Alice	30.000000	Los Angeles
2	103	Bob	27.166667	New York
3	104	John	22.000000	Unknown
4	105	Unknown	28.000000	Chicago
5	105	Bob	28.000000	Chicago
6	106	Eve	27.166667	Miami
7	107	Alice	30.000000	Los Angeles

In []: